Light – Reflection and Refraction

The big idea

We see objects because light from them reaches our eyes. Light travels in straight lines, **bounces off** surfaces (reflection) and **bends** when it passes from one medium to another (refraction). Mirrors and lenses use these two behaviours to form images.

Reflection and spherical mirrors

LAWS OF REFLECTION

1. The angle of incidence equals the angle of reflection.
2. The incident ray, the reflected ray and the normal all lie in the same plane.

A **concave** mirror curves inwards (like a spoon's inside); a **convex** mirror bulges outwards.

Key terms: **pole P** (centre of the mirror's surface), **centre of curvature C**, **radius of curvature R**, **principal axis**, **principal focus F**. For small apertures, **f = R/2**.

Object position (concave mirror)	Image
At infinity	At F; highly diminished, real, inverted
Beyond C	Between F and C; diminished, real, inverted
At C	At C; same size, real, inverted
Between C and F	Beyond C; enlarged, real, inverted
At F	At infinity; highly enlarged

Object position (concave mirror)	Image
Between P and F	Behind the mirror; enlarged, **virtual, erect**

A **convex** mirror always gives a **virtual, erect, diminished** image, so it gives a wide field of view: rear-view mirrors in vehicles. Concave mirrors are used in torches and headlights (to make parallel beams), by dentists and in shaving mirrors (enlarged erect image), and in solar furnaces.

MIRROR FORMULA AND MAGNIFICATION

$$1/v + 1/u = 1/f$$

$$m = h'/h = -v/u$$

Sign convention (New Cartesian): measure all distances from the pole; distances in the direction of the incident light are positive, against it negative. Heights above the axis are positive. The object is usually placed on the left, so **u is negative**. For a concave mirror f is negative; for a convex mirror f is positive.

AN OBJECT IS 30 CM IN FRONT OF A CONCAVE MIRROR OF FOCAL LENGTH 15 CM. FIND THE IMAGE.

$u = -30$, $f = -15$. $1/v = 1/f - 1/u = -1/15 + 1/30 = -1/30 \Rightarrow$ **v = −30 cm** (in front, real).

$m = -v/u = -(-30)/(-30) = -1$: **same size, inverted**. The object is at C.

Refraction

Light bends because it travels at different **speeds** in different media. Going from rarer to denser (air → glass), it bends **towards** the normal; denser to rarer, it bends **away**.

LAWS OF REFRACTION

1. The incident ray, refracted ray and normal lie in one plane.

2. **Snell's law**: sin i / sin r = constant = refractive index of medium 2 with respect to medium 1.

REFRACTIVE INDEX

n = speed of light in vacuum / speed of light in the medium = c/v

n of glass ≈ 1.5, water ≈ 1.33, diamond ≈ 2.42

Through a rectangular glass slab, the emergent ray is **parallel** to the incident ray but **laterally displaced**. A pencil in water looks bent and a pool looks shallower because of refraction.

Lenses

A **convex (converging)** lens is thicker in the middle; a **concave (diverging)** lens is thinner in the middle. A convex lens forms images much like a concave mirror (real for distant objects, virtual and enlarged when the object is between F and the optical centre: a magnifying glass). A concave lens always forms a **virtual, erect, diminished** image.

LENS FORMULA, MAGNIFICATION AND POWER

1/v − 1/u = 1/f

m = h′/h = v/u

P = 1/f (f in metres); unit dioptre (D). Convex lens: positive power. Concave lens: negative.

A CONCAVE LENS HAS FOCAL LENGTH 15 CM. WHERE SHOULD THE OBJECT BE PLACED SO THAT THE IMAGE IS 10 CM FROM THE LENS?

f = −15, v = −10 (virtual, same side). 1/u = 1/v − 1/f = −1/10 + 1/15 = −1/30 ⇒ **u = −30 cm**.

MIXING THE TWO FORMULAS

Mirror: $1/v + 1/u = 1/f$, $m = -v/u$. Lens: $1/v - 1/u = 1/f$, $m = +v/u$.

Quick recap

- Reflection: angle i = angle r. Concave mirror image depends on object position; convex always virtual and small.
- Refraction: $n = c/v$; light bends towards the normal entering a denser medium.
- Use the sign convention carefully with the mirror and lens formulas.
- Power $P = 1/f$ (metres), in dioptres.

Practice questions

1. A convex mirror is used as a rear-view mirror because it gives

   (a) a real, enlarged image (b) an erect, diminished image with a wide field of view (c) an inverted image (d) no image

2. The focal length of a concave mirror of radius of curvature 20 cm is

   (a) 10 cm (b) 20 cm (c) 40 cm (d) 5 cm

3. The refractive index of glass is 1.5. The speed of light in glass is ($c = 3 \times 10^8$ m/s)

   (a) 4.5×10^8 m/s (b) 2×10^8 m/s (c) 3×10^8 m/s (d) 1.5×10^8 m/s

4. The power of a convex lens of focal length 50 cm is

   (a) +0.5 D (b) +2 D (c) –2 D (d) +50 D

5. When light goes from air into water, it

   (a) speeds up and bends away from the normal (b) slows down and bends towards the normal (c) does not bend (d) reflects completely

6. A virtual, erect and enlarged image is formed by a concave mirror when the object is

   (a) beyond C (b) at C (c) at F (d) between P and F

Answers: 1. b 2. a 3. b 4. b 5. b 6. d